Spirit-Masters, Supernatural Punishment, and Religious Cooperation Among the Altai Uriankhai of Mongolia

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Abstract

Whether supernatural punishment beliefs promote cooperation has been studied extensively, yet much remains to be understood. For instance, most previous work has focused on "moralistic" religious traditions (e.g., Christianity, Hinduism), and how this may facilitate cooperation with distant co-religionists. Here, we use data from the Altai Uriankhai of Western Mongolia to examine whether punishment and omniscience beliefs regarding *local* deities (here, spirit-masters) promotes cooperation among distant co-religionists of this spirit-master tradition. Cooperation with distant co-religionists was assessed using two experimental games, the Random Allocation Game and Dictator Game. A formal causal inference approach was employed to specify our estimand of interest – i.e., the joint causal effect of punishment and omniscience beliefs on cooperation with distant co-religionists – and the assumptions required for a causal interpretation. Across most models a negative relationship between spirit-master beliefs and cooperation with distant co-religionists was observed, although many were also consistent with a null relationship. To the extent these results can be given a causal interpretation, they suggest that supernatural beliefs regarding local deities do not promote cooperation beyond parochial boundaries in this population, although whether the relationship is negative or null is equivocal.

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1. Background

1.1. Theory

Evolutionary theory draws our attention to a variety of problems inherent in the evolution of cooperation (Smith, 1982). Central to this issue is why individuals would cooperate with others when they can stand to benefit more by being selfish. The evolution of cooperative strategies thus requires mechanisms to ensure that cooperators interact with each other and that the benefits of cooperating outweigh the benefits of defection. Humans are exceptionally good at cooperating with each other, with many accounts for how and why this behaviour evolved. With respect to religion, one suggestion is that our ability to infer others' mental states contributes to our appreciation of the potential consequences of our social behaviors. However, anticipating others' behaviors by inferring their motivations runs the risk of exploitation. That is, having the ability to "mind-read" might turn exploitative without other mechanisms in place to uphold cooperation, such as effective mechanisms of punishment. The "supernatural punishment hypothesis" (Johnson & Krüger, 2004; Schloss & Murray, 2011) has been proposed as a potential explanation for human cooperation. While there are different variants of this general hypothesis (Schloss & Murray, 2011), one account posits that the fear of supernatural consequences will evolve under conditions where the costs of secular punishment and the chances of getting caught outweigh any benefits that could have been obtained by defecting in social dilemmas (Johnson & Bering, 2006). Although based on secular punishment, according to this account fear of supernatural punishment may alter both the perceived costs of secular punishment and the probability of getting caught, and hence promote cooperation. As such, in particular contexts, beliefs in supernatural punishment may represent one mechanism by which people can maintain cooperation.

Variants of this hypothesis attend to how far this effect can go (Norenzayan, 2013; Roes & Raymond, 2003). One view (Norenzayan et al., 2016) theorizes that as societies scale up through intergroup conflict, gods increasingly become more punitive, knowledgeable, and morally concerned. Through time, this auto-catalytic process boosts cooperation beyond parochial boundaries and expands gods' social relevance accordingly. Through conquest and conversion, this heightened prosociality may partly explain the now global ubiquity of the so-called "prosocial" or "moralistic" religions (e.g., Christianity and Islam). In this view, then, beliefs in gods' punishment, knowledge breadth, and moral concern should predict cooperation beyond local parochial boundaries, perhaps especially in more "moralistic" religious traditions.

Previous cross-cultural research using experimental economic games finds that the more individuals claim their gods punish and know, the more likely they are to play honestly and generously toward anonymous co-religionist strangers from geographically distant communities who are unlikely to ever reciprocate (Lang et al., 2019; Purzycki et al., 2016b). This work, however, was limited in a few critical ways that did not address some core aspects of the theory

(see Purzycki et al., 2022). First, the bulk of this work examined the association between supernatural punishment beliefs and cooperation among individuals in the "moralistic traditions." In other words, recipients were typically identified as being from the same "moralistic tradition" (e.g., Christianity, Hinduism, and others). As such, this project made no clear contrast between cooperation streams of "moralistic" traditions as compared to other religious traditions. Second, across a range of statistical model specifications, recent analyses found no effect of gods' moral concerns with cooperation (Bendixen et al., 2023). Across a multitude of model specifications and two focal predictors – a scale measuring gods' moral interests and free-list data content, gods' moral concerns – these moral concerns did not have a clear effect on cooperative outcomes to any reliable extent. This suggests that individuals' beliefs about supernatural punishment may shape cooperation to a greater extent than their perceptions about deities' moral concerns. Third, much previous research in this area did not take a causal approach to this question (Hernán & Robins, 2020; Pearl et al., 2016; although for an exception see Bendixen et al., 2023). The extent to which the previously reported associations reflect causal estimates may therefore be difficult to know with any degree of certainty.

Here, we build on this work by examining whether findings regarding supernatural punishment in "moralistic" religious traditions extend to "local" religious traditions, using a measure of "supernatural belief" which include both punishment and omniscience, while adopting an explicitly causal approach. The focal population of this report is the Altai Uriankhai of Mongolia. Based on the above theory, we anticipate the relationship between belief in punitive and knowledgeable gods with more fair and generous offers to geographically distant co-religionists who are unlikely to directly reciprocate to be either: i) positive, if the "supernatural punishment hypothesis" is more general and applies to local religious traditions; or ii) null, if the motivating effects of supernatural punishment are weaker or non-existent outside of "moralistic" religious traditions.

1.2. Ethnographic Background of Field Site

1.2.1. General background

The field site is Bosgo *bag* (third-level subdivision) of Duut *sum* (administrative unit of a province) in Khovd *aimag* (province), Western Mongolia. This site is located in a mountainous range, the eastern slopes of the east-central area of the Altai Mountains, which is sparsely inhabited by the Altai Uriankhai ethnic community, whose current population is about 3,700 people (in 2024) in two neighboring *sum*s, Duut and Mönkhkhairkhan, and in much fewer numbers in the neighboring Bayan-Ölgii *aimag* as well (Coloo, 1976; Lkhagvasuren, 2010). Roughly 1,500 Altai Uriankhai constitute more than 80% of the total population of Duut *sum*, whose center is Bosgo, 76 km southward from the province center Khovd city. According to the general belief among scholars of Mongolian studies (cf. Atwood, 2004; Birtalan, 2003), the Altai Uriankhais should share many cultural, ethnic, and historical ties with the neighbouring Tyvan

(endonym: Tyva, see Potanin, 1881) ethnic group in Russia, China, and Mongolia (cf. Mawkanuli, 1999). However, today most Altai Uriankhais (endonym: Uriankhai) emphasize their Mongolian origins and apparently deny any connection to Tyvans (Atwood, 2004; cf. Mongush, 1996), which corresponds well with their dominantly Oirat clan names (Dörvöd, Myangad, Torguud) and otherwise entirely Mongolian ones (Emch, Chono, Bor Ükhert, Darkhad, etc.), referred to among Oirats (Altai Uriankhai, Bayad, Dörvöd, Myangad, Ööld, Torguud, etc.) as *elken*.

In modern Mongolia, the term "Uriankhai" refers to a vaguely defined *yastan* (subethnic group; see Atwood, 2004; Ellis, 2024) in the western part of the country. The Altai Uriankhai form a distinct and coherent subgroup within this artificially constructed subethnic group, and have likely lived in the Altai region since the second half of the 17th century (Atwood, 2004; Lkhagvasuren, 2010).

The pastoral nomadic Altai Uriankhai communities live in a patriarchal (patrilineal) society with a patrilocal (virilocal) residence structure, consistent with all known Central Eurasian pastoral societies (Chaix et al., 2007; Krader, 1963). The earliest genetic evidence of patrilocal and patrilineal burial practices in Inner Asia has been identified in the Aldy-Bel and Sagly cultures from the territory of the modern neighbouring Tuva Republic, *ca.* 7th/6th century BCE (Mary et al., 2019). This social system appears to have remained continuous and uninterrupted for the past three millennia among Central Eurasian pastoral nomadic societies. Some extent of long-term cultural continuity has also been observed in the Altai Region (Jacobson-Tepfer, 2023).

1.2.2. Religious Landscape

While Mongolia is ethnically and geographically diverse, the country's primary religious tradition is a complex, syncretic synthesis of Tibetan Buddhism (or Lamaism, "the yellow faith"; see, e.g., Heissig, 1980; Sagaster, 2007) and the so-called Mongolian shamanism (see, e.g., Diószegi, 1961; Purev & Purvee, 2007; "the black faith," Banzarov, 1981), interwoven with spirit-master beliefs and ancestor worship. Since Lamaism persecuted shamanism during the sixteenth and seventeenth centuries, shamanistic beliefs and rituals were likely camouflaged by adopting Lamaist features (Heissig, 1980).

The previous nature of Mongolian shamanism can be reconstructed from shamanic hymns and prayers transmitted orally through the centuries and recorded during mid-twentieth-century field research – provided that elements derived from Lamaism, such as names and conceptions of gods, can be distinguished and set aside. At the same time, Buddhism in Mongolia absorbed many features of shamanism (Atwood, 1996; Heissig, 1980; Pegg, 2001), which appears to have originated in ancestor worship (Heissig, 1980).

The data gained in the present fieldwork fits with that of the historical reports, strongly attending to the local spirit-master beliefs. The majority of Altai Uriankhais' beliefs and ritual practices

exhibit relatively limited impacts of Lamaism. Almost every participant denoted several local spirits, such as the Lord/Spirit/Deity (*savdag*) of the Altai (Mountains), the Spirits of land and water ("nature"; *lus savdag*), *Tenger* ("Heaven")/*Mönkh khökh tenger* ("Eternal Blue Sky"; cf. Pegg, 2001), the spirit of Mother and Father (cf. Ancestor worship), *Doloon burkhan od* (the Big Dipper; "Great Bear"/Ursa Major; cf. Banzarov, 1981; Heissig, 1980), the Mountains and Rivers as helping spirits (e.g., *Baatar-Khairkhan*, *Altai Khangai*, *Khongor Khairkhan* (all mountains), *Tsenkheriin gol* (a river), *Tsagaan khad* (a rock), *Gurvan tsenkheriin sakhius* (a protective deity of a rock cave), etc.

Milk or milk tea (*süütei tsai*) offerings (libations) and incense offerings are also common practices of pre-Buddhist origin among religious Mongolians (Heissig, 1980). Milk is widely regarded as a symbol of purity and positive energy. In open spaces, it is customary to sprinkle milk or milk tea before anyone consumes it – a ritual that is practiced independently of Buddhism. During these offerings to the heavens, individuals concentrate solely on positive thoughts, expressing wishes for good fortune, health, longevity, protection from natural disasters, safe travels, and the avoidance of misfortune. The act of sprinkling also includes a respectful offering to the spirits of one's ancestors.

There is a significant difference between the frequency of Buddhist and spirit-master rituals among the Altai Uriankhai. They perform significantly fewer rituals to Buddha than to local spirit-masters. Of the 86 participants who took part in the Religiosity Interview (described in the Methods below), 57 (66%) perform rituals to the Buddha with varying frequency, while 29 (34%) do not. In contrast, 79 (92%) out of the 86 informants perform rituals to spirit-masters with varying frequency, 6 do not, and one informant was unsure.

Other likely animistic and shamanistic religious traditions dating back to pre-Buddhist times include the worship of fire (fire cult) and the celebration of *Tsagaan Sar* ("White Month"), the Mongolian Lunar New Year, and the use of the lunar calendar (Atwood, 1996; Banzarov, 1981; Heissig, 1980). In the daily life of nomadic communities, fire holds a central and sacred role. Each household deeply respects its own fire, treating it with the utmost reverence. Certain substances, such as blood, mucus, and similar materials, are strictly forbidden from being burned in the fire, as doing so is believed to invite misfortune. To preserve the sanctity of the fire, specific chants and rituals are performed. Additionally, it is taboo to use another family's fire pot, especially the one designated solely for the household fire. It is also believed that children under the age of three can perceive spiritual entities within the flames. The nomadic Mongolian lifestyle is closely intertwined with *Tsagaan Sar* (the White Month or Lunar New Year). Herders regard *Tsagaan Sar* as a symbol of positivity, as it marks the arrival of the warm season and the birth of young animals, signaling the end of the harsh winter. Given that rural households often live at considerable distances from one another, *Tsagaan Sar* serves as an important occasion for social greetings and reunions, typically occurring at least once annually. The recitation of Buddhist sutras (mentioned by 3/20 participants as part of the Religious Landscape Interview

[RLI; see Methods below]) by a lama can be considered among the less syncretic Buddhist elements of local ritual tradition. Although traditionally not conceived as such by religious scholars, throughout Mongolia and beyond, many consider Buddha (Khalkha Mongolian: [*Budda*] *burkhan*) to be a punitive, omniscient, and morally concerned deity (Berniūnas et al., 2020; Purzycki & Holland, 2019).

Religious expression takes many forms, but we focus here on practices dedicated to local spirit-masters at ritual cairns, or *ovoo* (depending on dialect, variously referred to across the wider region as *ovaa, obo,* or *oboo*; cf. Anderson & Harrison, 2003; Coloo, 1988). *Ovoo* offerings are a ritual expression of reverence toward *Tenger* (the sky or heavens). Their primary purpose is to seek blessings for good fortune, including protection from natural calamities such as heavy snowfall, drought, and other environmental hardships, as well as to pray for longevity and good health. Prior to hunting, people often appeal to *Tenger* for *hishig*, a term that signifies divine favor, particularly in the form of successful hunting and sufficient food. Offerings of milk and milk tea are commonly made at *ovoo*, in addition to local food customs. These typically include "white products" (such as milk and cheese), sweets (candies and cakes), "black products" (primarily alcoholic beverages, usually vodka in Russia and Mongolia), and "red products" (such as lamb, beef, or horse meat, depending on the deity's preference). Other common offerings include lumps of butter, pure water, fruit, and, in some cases, banknotes (Dumont, 2024).

In a few important ways, *ovoo* piety also bears relevance to the potential expansion of more reliable cooperation. First, ethnographic observations suggest that ritual cairn participation can be a reciprocal affair across disparate families (Mongush, 1992). This is especially pertinent, given that the tradition may have developed in part to mediate territorial boundaries between non-kin under conditions of increased population densities (Purzycki, 2010). Second, previous research suggests that observing others' participation in cairn rites might bolster their perceived trustworthiness (Purzycki & Arakchaa, 2013). Third, the sheer breadth of geographical coverage of this tradition suggests a particular utility of unification among Inner Asians. Fourth, the global, culturally independent emergence of similar traditions suggests a particular utility under similar conditions (Sierksma, 1963).

2. Methods

With a few minor updates, we followed the standards, methods, and protocols from the *Evolution of Religion and Morality* (ERM) *Project* (Lang et al., 2023; Purzycki et al., 2016a, 2024). In this case, rather than use preliminary ethnographic interviews to select deities to ask about, we selected the deities–Buddha (*Budda burkhan*) and spirit-masters (*uul usnii ezed;* lit. "spirits of the Mountains and Rivers") based on our prior knowledge of the religious landscape in our field sites, our interest in these particular deities, and the comparison afforded by previous research conducted in a nearby region in Tyva, southern Siberia (Purzycki & Kulundary, 2018). We translated all original English materials into Standard (Khalkha) Mongolian, back-translated, and subsequently edited for clarity (available here on the project GitHub page: https://github.com/bgpurzycki/GGSL-Project/tree/main/Protocols/Objective%201).

2.1. Theoretical Model

Our estimand (i.e., target of inference) is the causal effect of supernatural beliefs regarding spirit-masters on cooperation with distant co-religionists (*ovoo* practitioners) in two experimental games (the Random Allocation Game [RAG] and Dictator Game [DG]). Based on our assumed data-generating mechanism (see the Directed Acyclic Graph [DAG] in Figure 1), the following covariates are core confounders we need to control for to estimate an unbiased causal effect: age, sex, education, material/food security and number of children (this model was adapted from our core causal model for the wider project, available: https://github.com/bgpurzycki/GGSL-Project/blob/main/Pre-Registration/Objective%201%20-%20Core%20Model/GGSL_Obj1_Core_Causal_Model.pdf). The rationale for this is that each may plausibly cause both religiosity/supernatural punishment and cooperation (Baimel et al., 2022; Inglehart, 2021; Vardy et al., 2022), and hence result in bias if not adjusted for in analyses. While not confounders, we also include features of the game structure which may impact levels of cooperation (game order, comprehension checks) as they may help improve the precision of our estimates (and, at the very least, will not introduce any bias).

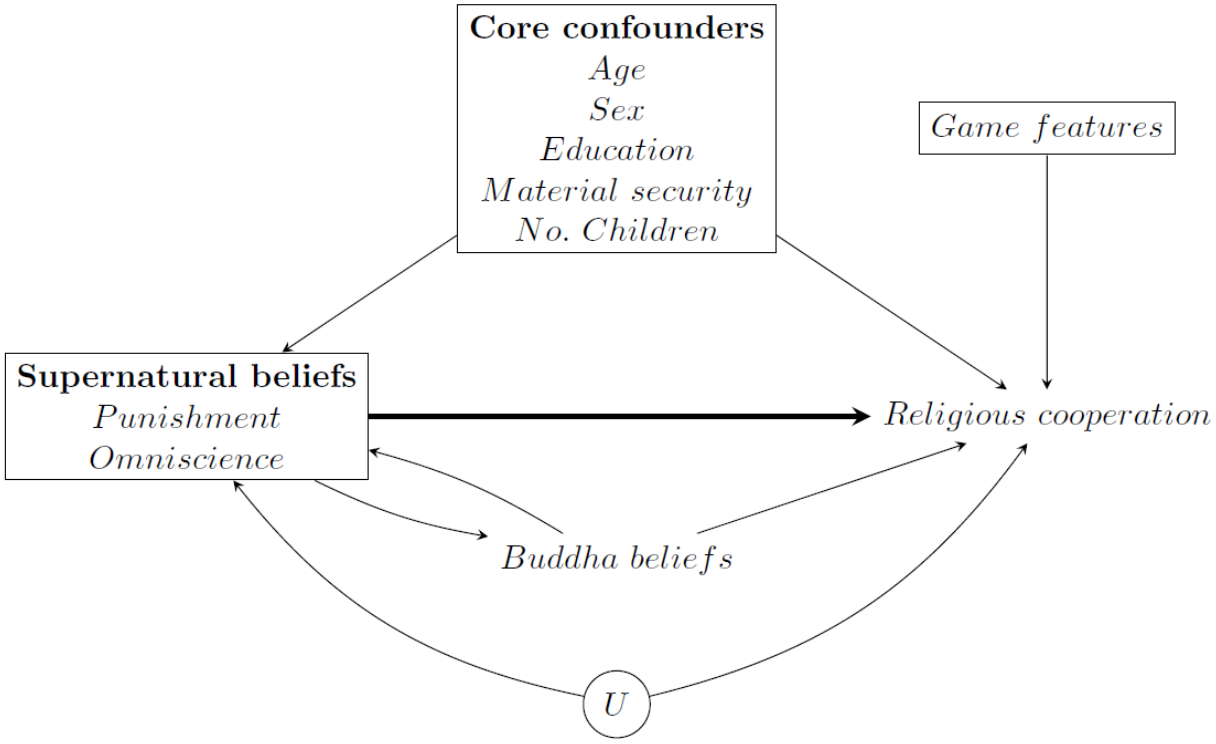


Figure 1: Directed Acyclic Graph (DAG) summarising our causal assumptions regarding the relationship between supernatural beliefs regarding spirit-masters and cooperation with distant co-religionists. The bold arrow denotes the estimand of interest. Note that causal relations between the core confounders are not displayed here. Causal relations between the components of supernatural beliefs are also not displayed here, as our estimand is the joint effect of supernatural beliefs on religious cooperation (see the main text for more discussion on this). The set *U* refers to potential unmeasured confounders.

In addition to this core set of confounders, we will also consider the role of supernatural beliefs regarding an alternative deity, Buddha. Previous research using the ERM data has shown that – across a range of societies – beliefs about local gods are somewhat independent from moralistic gods (Purzycki, Willard, et al., 2022). However, disentangling the complex causal webs between these beliefs is difficult. For instance, perhaps beliefs about Buddha – as they are from a larger "world" religion – impact beliefs about spirit-masters (in which case these Buddha beliefs would be confounders, and need adjusting for in order to estimate an unbiased causal effect). Conversely, individuals who strongly hold beliefs about spirit-masters may believe less in Buddhism (in which case beliefs about Buddha may be a mediator between spirit-master beliefs and cooperation, and hence adjustment for Buddha beliefs may result in overadjustment bias). Alternatively, perhaps beliefs about Buddha and spirit-masters come from a shared, unobserved, latent 'religiosity' or other cognitive factor (in which case adjustment for Buddha beliefs would be again necessary to block this back-door confounding path). On the other hand, it could be that Buddha and spirit-master beliefs are independent of one another (and hence no need to adjust for

Buddha beliefs). It may also be that Buddha and spirit-master beliefs are practically identical (meaning they are not possible to separate). As a final – but non-exhaustive – scenario, perhaps Buddha and spirit-master beliefs continually reinforce and shape one another (in which case both adjusting and not adjusting for Buddha beliefs will lead to bias). As such, given this complexity we will conduct sensitivity analyses comparing results with vs. without Buddha beliefs as a covariate to assess whether these different assumptions impact our inferences (assuming Buddha and spirit-master beliefs are not completely overlapping, as assessed via descriptive statistics).

Our exposure variables are a range of supernatural beliefs about spirit-masters, based on two broad categories: punishment and monitoring/omniscience (described in more detail below). Due to the difficulties in separating the causal effects of these religious belief components (e.g., are they independent? Do they reciprocally cause one another? Are they caused by a latent unobserved "religiosity" factor?), we decided to focus on the joint effect of both religious belief variables together. That is, our focal contrast will be when punishment and omniscience both take the value 0 vs. when punishment and omniscience both take the value 1. This means that we cannot isolate the causal effect of, say, supernatural punishment, by itself, but we do assume that we can measure the joint causal effect of these variables together (hence why causal relations between these individual religious belief components are not modelled in the DAG above).

Although punishment and omniscience mini-scales have been used previously, it is possible that the punishment (and similarly for omniscience) variables do not measure precisely the same thing. For instance, the god's influence in an afterlife (one variable of the punishment mini-scale) could reflect supernatural rewards, in addition to supernatural punishment (it is also possible that the god may only punish living people, with no reach in the afterlife); similarly, for omniscience, spirit-masters may only care/know about the behaviours of local people, not those of people far away (i.e., a local, rather than universal, god). We will therefore repeat analyses using just single punishment and omniscience items – whether spirit-masters ever punish, and whether spirit-masters can see peoples' thoughts and feelings, respectively – to assess whether we obtain similar results (see below for additional details of these variables).

Our theoretical model assumes no unmeasured confounding, no measurement error or residual confounding, that our causal model is correctly specified, and that our sample is broadly representative of our target population (i.e., wider Altai Uriankhais). As our aim is causal inference, we have endeavoured to make our causal assumptions clear; we will return to whether these assumptions are plausible in the Discussion section.

2.2. Materials and Procedure

Initial data collection took place in Summer 2024. During the first two days of fieldwork, we conducted a preliminary Religious Landscape Interview (RLI) with 20 participants – 12 in Khovd city and 8 in Bosgo. The aim of this initial interview was to better understand the

religious context, providing ethnographic insight when conducting and interpreting subsequent quantitative data collection. These individuals were not included in the subsequent experiments. We then administered a Demography Interview (26 questions) and a Religiosity Interview (62 questions) to all 88 participants who took part in the experimental cooperative games. The Religiosity Interview included questions about general religiosity, the designated "moralistic" god (Buddha), and "local" god (spirit-masters), and participants' thoughts on others.

The interviews and the experiments were primarily conducted in a two-room outbuilding of a private home in Bosgo (80 of the total 88). A smaller number of sessions were held in classrooms at Khovd State University in Khovd city (8 participants). Note that due to errors in data collection of the scale questions in the Religiosity Interview, this questionnaire was asked again to as many of the same participants as possible in the next field season in Summer 2025; however, these errors do not affect any of the variables used here, hence we focus on the data collected in 2024.

2.2.1. Demographics

We asked a wide range of questions pertaining to demography. As already mentioned, of particular importance here are participant age, sex, years of formal education, food insecurity, and number of children which we model as confounders between beliefs and behavioral outcomes.

2.2.2. Religiosity variables

Our supernatural beliefs variables of interest consist of:

1. Punishment: Based on the questions "Do spirit-masters ever punish people for their behavior?" (yes = 1, no = 0) and "Can spirit-masters influence what happens to people after they die?" (yes = 1, no = 0). As per previous ERM work (Purzycki et al., 2016b), these two variables will be averaged together. These variables were averaged for both theoretical reasons (as they likely measure the same/similar constructs) and practical reasons (to limit the number of interaction terms; see below).
2. Omniscience: Based on the questions "Can spirit-masters see into people's hearts or know their thoughts and feelings?" (yes = 1, no = 0) and "Can spirit-masters see what people are doing if they are far away?" (yes = 1, no = 0; note that, for consistency with previous ERM data and other field sites in the project, this question should have been in reference to a specific known but distant town/city, but was not here). As above, answers to these two questions were averaged together.

We focus on punishment and omniscience as these are key supernatural beliefs which, on both empirical and theoretical grounds (Lang et al., 2019; Purzycki et al., 2016b, 2018), may promote cooperation among distant co-religionists. Note that we do not explore "moral concerns" here for a number of overlapping pragmatic and conceptual reasons, including: gods' moral concern does

not appear to motivate cooperative behaviour (Bendixen et al., 2023; Lang et al., 2019; Purzycki et al., 2016b), the measurement (and hence interpretation) of questions regarding "moral concern" is fraught (i.e., different measures of moral concern are not correlated, suggesting they measure different things; Bendixen & Purzycki, 2025), and there were specific issues in Altai Uriankhai with the collection of "moral concern" Likert scale questions.

2.2.3. Behavioral experiments

To measure cooperative outcomes, we deployed two variants of the Random Allocation Game (RAG; Hruschka et al., 2014) and Dictator Game (DG). In each game type, participants allocated money in two dyads: 1) one cup was reserved for themselves and one was for an anonymous *ovoo* practitioner from Bayandun *sum*, Dornod *aimag*, a district over 2,000 km away in Eastern Mongolia (SELF vs. DISTANT condition), and 2) an anonymous local co-religionist *ovoo* practitioner from Duut *sum* and another anonymous *ovoo* practitioner from Bayandun *sum* (LOCAL vs. DISTANT condition). Participants first played the RAGs in counterbalanced order, followed by the DGs in counterbalanced order. Recipients were denoted by opaque cups with labels on them. Participants used tokens each worth 1,000 MNT (~$0.30 USD).

In the RAGs, we instructed participants to think of which cup they would like to put a coin into and then roll a fair, 2-colored die. If the die came up black, then they should put the token into the cup they thought of, but if the die came up white, they were supposed to put the token into the cup opposite to what they thought. As participants played alone, they could bias their allocation to favor one recipient or the other, and thus violate the 50% chance of a coin going into any given cup. Each RAG entailed 30 die-rolls. For each DG variant, participants allocated up to 10 tokens. Using pre-tests, we ensured that players understood the rules before playing and knew that they would receive any money allocated to their own cups along with their show-up fee of 20,000 MNT (~$6 USD).

2.3. Statistical Model

Following our DAG, our exposure is the joint effect of religious beliefs of spirit-masters (punishment and omniscience), and our outcomes are RAG and DG (both SELF and LOCAL, with DISTANT as target). As detailed above, our core set of confounders includes: age, sex, education, material/food security and number of children. We used aggregated binomial models for RAGs, and ordinal models for DGs, with separate models for SELF vs. DISTANT and LOCAL vs. DISTANT. We employed Bayesian models using Stan via the package "brms" (Bürkner, 2017), with four chains with 2,000 iterations (1,000 chains as warm-up). Visual inspection of convergence between the chains, r-hat values close to 1.00 and adequate effective sample sizes were used to ensure that chains are well-mixed and sufficiently-powered. We used weakly regularising priors. Analyses were conducted R version 4.4.1 (R Core Team, 2024).

We used *g*-computation methods (Hernán & Robins, 2020) to estimate the marginal causal effect of a change in exposures (from both 0 to both 1). For interpretability, these marginal effects reflect the average change in the outcome (i.e., average treatment effect) on the count scale (e.g., the difference in the number of donations to DISTANT, for a change in supernatural punishment) based on posterior predictions from the models.

The exposures were first treated as additive (i.e., independent). To test whether a multiplicative model fits the data better, we then observed whether the inclusion of an interaction term between punishment and omniscience improved model fit via LOOIC (see Bendixen et al., 2023). We performed both of these model specifications as theory regarding supernatural punishment is under-specified and does not make clear predictions regarding the effects of supernatural beliefs on cooperation, but is plausible that there may be synergistic effects of these religious beliefs; for instance, believing in supernatural punishment *and* omniscience might have a larger effect than either when treated independently. However, we note that, due to a relatively small sample size (~90), our analyses may be underpowered to detect said interaction effect (Blake & Gangestad, 2020).

Because of small amounts of missing data, particularly in the religiosity variables, to boost statistical efficiency we used multiple imputation to create 20 imputed datasets. This number was chosen to broadly match the overall proportion of missing data in our focal complete-case analyses (approx. 25% missing) and generate sufficient variation in our imputed values (White et al., 2011), while also minimising computational time. All exposures, outcomes, confounders and covariates detailed above were included in our imputation model, in addition to auxiliary variables to predict missing religiosity data (participant believes in, ever thinks about, and ever performs rituals towards, deity, for both spirit-masters and Buddha). Given the relatively low levels of missing data per variable, when combined with the large amounts of additional auxiliary information to help impute missing data, we believe that our imputations meet the "Missing-At-Random" assumption required for unbiased imputations (Hughes et al., 2019; van Buuren, 2018; White et al., 2011), although we cannot rule out data being "Missing-Not-At-Random" and hence biased. Multiple imputation was performed using the "mice" R package (van Buuren, 2018). We used a burn-in period of 10 iterations per imputation. These imputed datasets were then analysed using the "brm_multiple" command in the package "brms" to run models on each imputed dataset. We used multiple imputation rather than Bayesian imputation in Stan, as multiple imputation is more flexible and easier to both handle missing categorical variables and incorporate auxiliary variables. While the imputed results form our main analyses, complete-case analyses will also be performed and results compared. Note that reported LOOIC statistics for the imputed data have been averaged across all imputed datasets.

In addition to these main analyses, we also conducted additional sensitivity analyses, as described above, including: i) whether the addition of Buddha beliefs as additional covariates

impacts results; and ii) using individual variables of punishment and omniscience, rather than mini-scales of composite variables.

Example scripts demonstrating the pre-registered analyses on simulated data are available here: https://github.com/bgpurzycki/GGSL-Project/tree/main/Pre-Registration/Objective%201%20-%20Altai%20Uriankhai.

2.4. Sampling and Participants

The sample was drawn from the Altai Uriankhai community in Bosgo *bag*, Duut *sum*, Khovd *aimag*, Mongolia. Using Mongolian population census data (National Statistics Office of Mongolia, 2021) and the online database of the National Statistics Office of Mongolia (National Statistics Office of Mongolia, 2025), we obtained a broadly representative sample in terms of age, sex, community size and birthplace. The sample was primarily composed of members of the Altai Uriankhai community living in or near Bosgo within Duut *sum*, with a smaller number (8 of 88) of participants being Altai Uriankhai who currently reside in Khovd city but were born and raised in Bosgo. No individuals declined to participate. The total number of participants was 88, with 38 males (43% of the sample). The mean age of male participants was 41.4 years (range 18 to 68), and 43.6 years for female participants (range 19 to 76). The predominant occupation among participants was herding: 19 of 38 males (50%) and 17 of 50 females (34%) were identified as herders, comprising 41% of the total sample (36 out of 88). Note that, although 88 participants took part in the initial demographic interviews, two participants did not complete the subsequent religiosity interview and experimental games; these participants have been excluded from the analyses reported below.

3.5. Pre-registration

A pre-registration for this paper can be found here: https://github.com/bgpurzycki/GGSL-Project/blob/main/Pre-Registration/Objective%201%20-%20Altai%20Uriankhai/GGSL_Altai%20Uriankhai_PreReg.pdf. Note that data for this study had already been collected, cleaned and processed ready for analysis prior to this pre-registration. Other than the deities selected, the methods protocol, however, was already designed prior to data collection and a part of the project's grant proposal, available here: https://github.com/bgpurzycki/GGSL-Project/blob/main/Pre-Registration/Proposal/TRT_Purzycki.pdf. When preparing the pre-registration we therefore had some information about the data and the specific ethnographic context, which was useful to help understand the complexities of the data (e.g., data collection irregularities regarding scale questions, as noted above) and help inform our causal model and statistical analyses (e.g., levels of missing data to inform imputations, contextual information regarding the relation between Buddhist and spirit-master beliefs). To be fully transparent, descriptive statistics for key variables and regressions based on the core causal model for the wider project (as noted above) were conducted in January 2026 for a project workshop prior to posting the pre-registration;

however, this was *after* the causal model outlined here had been finalized, meaning that this information did not influence our causal model or corresponding statistical approach outlined in the pre-registration.

The only deviation from our pre-registered analysis strategy was that "comprehension checks" were not included as a covariate in models with RAG donations as the outcome as there was no variation in comprehension in this sample. Additionally, while not a deviation from our pre-registration, we have provided more details about the multiple imputation procedure.

3. Results

3.1. Descriptive Statistics

3.1.1. Sociodemographics, Religious Beliefs, and Missing Data

A summary of the sociodemographic and religiosity data is presented in Table 1, for both raw and imputed data, along with patterns of missing data. There was little missingness in the sociodemographic data, but more in the religiosity data, especially for Buddha punishment and omniscience beliefs (up to 25% missing). These missing data are primarily because many of these participants did not believe in Buddha and answered "don't know" to these questions (approximately one-quarter of participants claimed not to believe in Buddha, compared to only two participants who claimed the same for spirit-masters). The use of auxiliary variables of belief in, performing rituals towards, and ever thinking about these deities would help when imputing these missing values. As such, these beliefs towards Buddha were much lower in the imputed samples compared to the raw data. Analyses relying on complete-case data are therefore biased towards individuals who believe in Buddha and their punitive and omniscience capabilities, meaning that complete-case analyses may be at risk of selection bias if these beliefs are related to both our exposure and outcome.

Focusing on the religiosity data, almost all participants believed that spirit-masters punished people for their behaviour (94.2%), which was higher than corresponding beliefs in Buddha punishment (raw = 79.7%; imputed = 66.9%). Fewer participants believed spirit-masters could impact what happens to people after they die (raw = 69.4%; imputed = 66.0%), similar to Buddha (raw = 75.0%; imputed = 61.8%). There were moderate-to-strong positive correlations between all punishment beliefs within and between deities (spirit-masters vs. Buddha punish: raw = 0.68; imputed = 0.56: spirit-masters vs. Buddha influence afterlife: raw = 0.77; imputed = 0.39; spirit-masters punish vs. influence afterlife: raw = 0.61; imputed = 0.56; Buddha punishes vs. influences afterlife: raw = 0.58; imputed = 0.69; all tetrachoric correlations). The mean of the punishment mini-scales were similar for both spirit-masters and Buddha in the raw data, albeit marginally higher for spirit-masters (0.81 vs. 0.78), although there was a larger difference in the imputed data (0.80 vs. 0.64). There was a strong correlation between these mini-scales in the raw data (polychoric correlation = 0.64), but this was weaker in the imputed data (0.33).

Regarding omniscience beliefs, a substantial majority of participants believed that spirit-masters could both know peoples' thoughts and feelings (raw = 84.6%; imputed = 82.0%) and see what people do far away (raw = 81.7%; imputed = 79.6%). These percentages were similar, but slightly lower, for the corresponding Buddha beliefs, especially in the imputed data (know thoughts and feelings: raw = 74.6%; imputed = 62.2%: see far away: raw = 79.7%; imputed = 66.6%). There were strong correlations between all these omniscience beliefs within and between deities (spirit-masters vs. Buddha know thoughts/feelings: raw = 0.70; imputed = 0.55: spirit-masters vs. Buddha see far away: raw = 0.81; imputed = 0.63: Spirit-masters know thoughts/feelings vs. see far away: raw = 0.68; imputed = 0.71: Buddha knows thoughts/feelings

vs. sees far away: raw = 0.92; imputed = 0.92: all tetrachoric correlations). As such, the mean of the omniscience mini-scales was higher for spirit-masters compared to Buddha, which was larger in the imputed data (raw = 0.84 vs. 0.77; imputed = 0.81 vs. 0.65). There was a strong correlation between these omniscience mini-scales (polychoric correlation in raw = 0.74: imputed = 0.61). There were also strong correlations between the punishment and omniscience mini-scales for both spirit-masters (raw = 0.67; imputed = 0.70) and Buddha (raw = 0.77; imputed = 0.83: all polychoric correlations).

Table 1: Summary of Altai Uriankhai sociodemographics and religious beliefs regarding punishment and omniscience towards both spirit-masters (local god) and Buddha (moralistic god; *n* = 86). Proportions of missing data for each variable are also presented, along with mean values across the 20 imputed datasets (for variables with any missing data). Note that the cell counts for imputed data can include decimal places, as they are averages.

Variable	N	Raw data - N (%) or median (IQR)	Imputed data - N (%) or median (IQR)	Missing data - N (%)
Age (years)	86	44.5 (31 to 50.75)	NA	0 (0.0%)
Sex (male)	86	36 (41.9%)	NA	0 (0.0%)
Education (years)	84	10 (8 to 14)	10 (8 to 14)	2 (2.3%)
Number of children	85	3 (2 to 4)	3 (2 to 4)	1 (1.2%)
Material security	83	0 (0 to 0)	0 (0 to 0)	3 (3.5%)
Spirit-masters ever punish behaviour	86	81 (94.2%)	NA	0 (0.0%)
Spirit-masters influence people after they die	72	50 (69.4%)	56.8 (66.0%)	14 (16.3%)
Spirit-masters know thoughts and feelings	78	66 (84.6%)	70.7 (82.2%)	8 (9.3%)
Spirit-masters can see people far away	82	67 (81.7%)	68.3 (79.4%)	4 (4.7%)
Buddha ever punishes behaviour	69	55 (79.7%)	57.5 (66.9%)	17 (19.8%)
Buddha influences people after they die	64	48 (75.0%)	53.2 (61.8%)	22 (25.6%)
Buddha knows thoughts and feelings	67	50 (74.6%)	53.8 (62.5%)	19 (22.1%)
Buddha can see people far away	69	55 (79.7%)	57.3 (66.6%)	17 (19.8%)

3.1.2. RAGs and DGs

There were no missing data in any of our game variables. For the local RAG, out of 30 allocations, participants gave on average 14.5 tokens (SD = 3.2) to distant *ovoo* co-religionists. Donations were slightly more biased to self in the self RAG, with a mean of 13.9 tokens (SD = 3.1) to distant co-religionists (Figure 2). Allocations in the local RAG were somewhat biased towards local co-religionists, although formal testing against a null binomial distribution using an intercept-only Bayesian regression model suggested equivocal evidence for bias (odds ratio = 0.94, 95% credible intervals [CI] = 0.87 to 1.01). The evidence for biased allocations was stronger in the self RAG condition (odds ratio = 0.86, 95% CI = 0.80 to 0.93), although there was only weak evidence that allocations differed between local vs. self conditions using a repeated-measured model (difference on count scale = -0.62, 95% CI = -1.60 to 0.37).

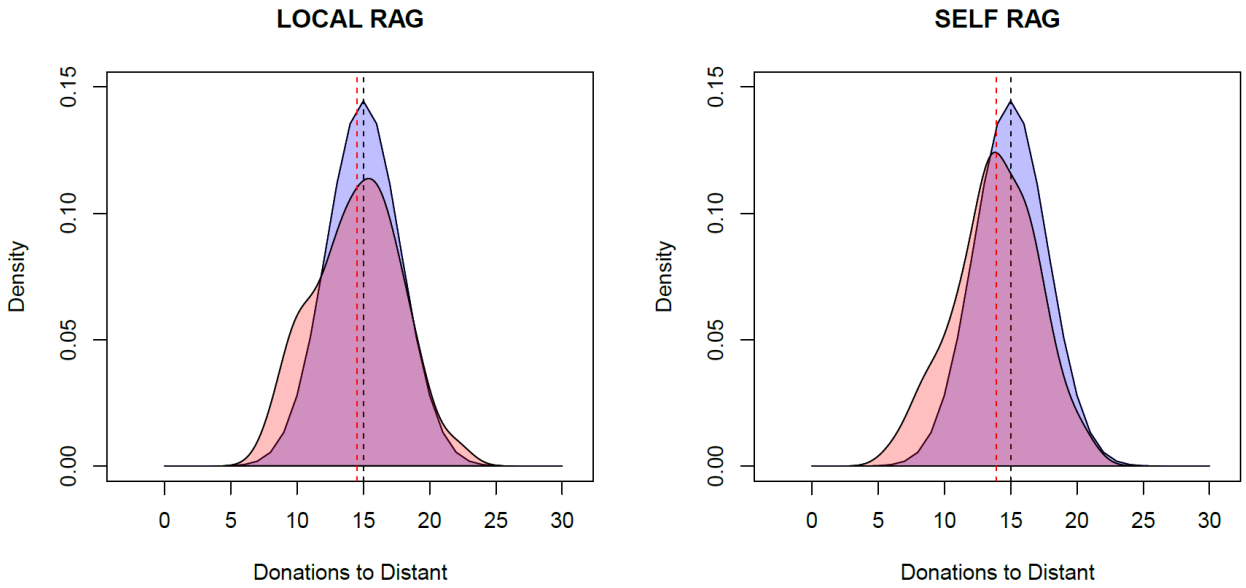


Figure 2: Random Allocation Game (RAG) donations to distant co-religionists in both the local (left) and self (right) conditions. The observed distribution is shaded in red, with the mean observed values indicated by the dashed red line; the theoretical binomial distribution assuming unbiased allocations is shaded in blue, with the mean expected theoretical value indicated by the dashed black line at 15 donations.

For the DGs, participants gave on average 4.6 donations (SD = 1.5) to distant co-religionists in the local DG condition, which was similar in the self DG condition (M = 4.7, SD = 1.4; Figure 3). In a repeated-measures model, there was no evidence for a difference in donations between conditions (difference on count scale = 0.15, 95% CI = -0.37 to 0.66).

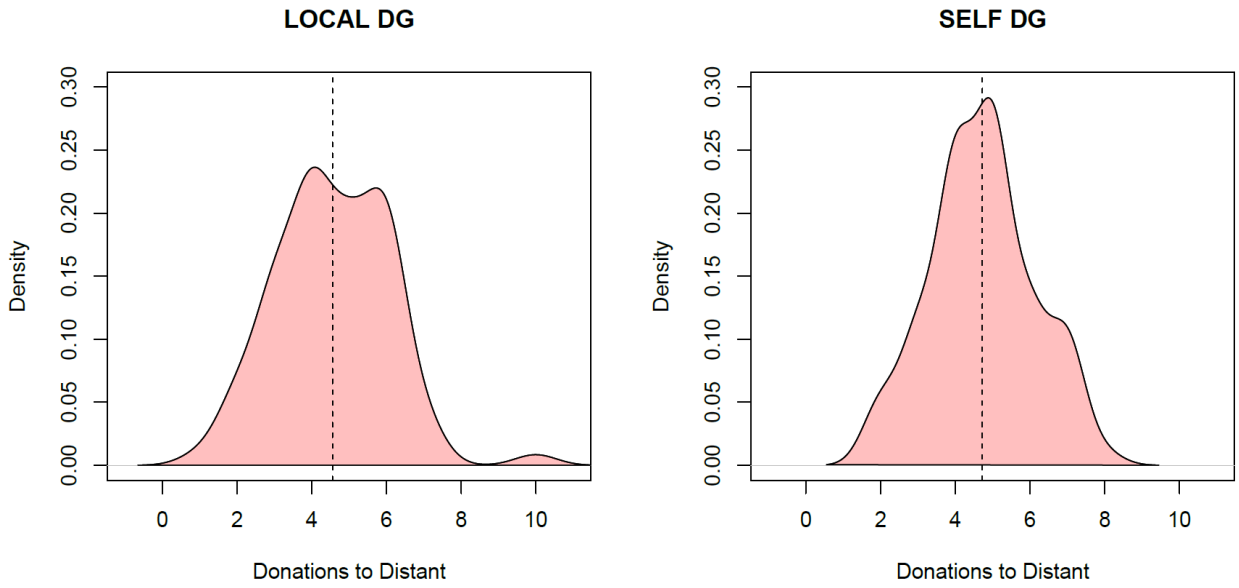


Figure 3: Dictator Game (DG) donations to distant co-religionists in both the local (left) and self (right) conditions. The observed distribution is shaded in red, with the mean observed values indicated by the dashed line.

3.2. RAG and DG Results

Next, we describe the results of our regression and *g*-computation analyses, based on the causal assumptions presented above (Figure 1). Our main results focus on the analyses using multiply-imputed data (n = 86), with supplementary results using the complete-case sample (i.e., participants with complete exposure, outcome and covariate data across all model specifications; n = 49; 43.0% missing). A summary of all results is presented in Table 2.

3.2.1. RAG Results

Results are presented first with the "core confounders" of Figure 1 as assumed confounders of our focal estimand (i.e., age, sex, education, number of children and material security). For the local RAG condition, the inclusion of an interaction term between spirit-master belief exposures did not improve model fit (ELPD LOO for additive model = -228.8; ELPD LOO for interaction model = -229.8; ELPD diff = -1.0, SE diff = 0.5). Based on *g*-computation results from the additive model, there was evidence for a slight decrease in the number of tokens – by approximately one – to distant *ovoo* co-religionists with increasing belief in punitive and omniscient spirit-masters, although the range of estimates was also consistent with no relationship (median estimated difference in number of tokens to distant = -1.00, 95% CI = -3.50 to 1.48; from approximately 15.3 tokens when both exposures took the value '0' to 14.3 when both were '1'; Figure 4).

In the self RAG, there was again no evidence for a multiplicative effect of the exposure on allocations (ELPD LOO for additive model = -225.0; ELPD LOO for interaction model = -225.8; ELPD diff = -0.8, SE diff = 0.4). The joint causal effect of spirit-master beliefs was in the same

negative direction as for the local RAG, but with stronger evidence for a relationship, with approximately three fewer tokens given to distant co-religionists, on average (Figure 4; median diff = -3.33, 95% CI = -5.79 to -0.76; from approximately 16.6 tokens when both punishment and omniscience were '0' to 13.2 tokens when both were '1').

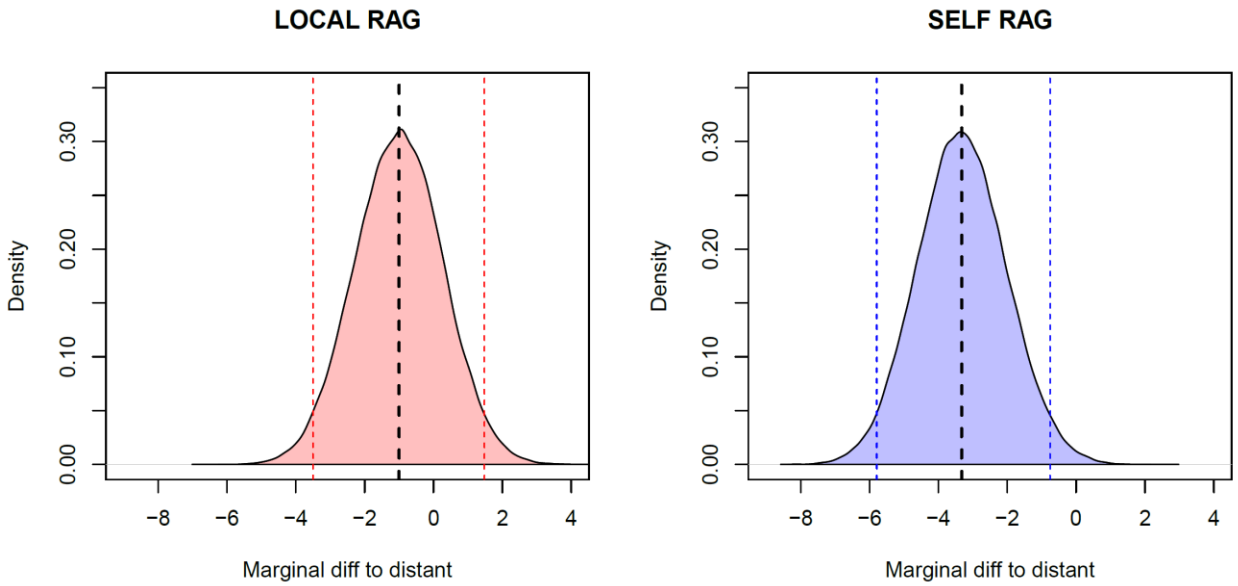


Figure 4: Marginal effect of change in spirit-master punishment and omniscience beliefs on change in cooperation with distant *ovoo* practitioner co-religionist in the Random Allocation Games (RAGs; local RAG = left; self RAG = right). Dashed black lines denote the median expected difference in cooperation (number of tokens) from both punishment and omniscience taking the value "0" to both as "1," with 95% credible intervals in lighter dashed lines.

3.2.2. DG Results

Again focusing first on the model with core confounders, an interaction term between exposures did not improve model fit for the local DG model (ELPD LOO for additive model = -159.7; ELPD LOO for interaction model = -159.6; ELPD diff = -0.1, SE diff = 0.4). Based on this additive model, there was again evidence for a weak negative relationship between spirit-master beliefs and cooperation with distant co-religionists, although the range of plausible estimates included zero (Figure 5; median diff = -0.70, 95% CI = -1.92 to 0.49; from approximately 5.1 tokens when both punishment and omniscience were '0' to 4.4 tokens when both were '1').

For the self DG, there was again no improvement in model fit when assuming a non-additive relationship between the exposures (ELPD LOO for additive model = -155.1; ELPD LOO for interaction model = -155.1; ELPD diff = 0.0, SE diff = 0.3). As with the local DG results above, the majority of the probability mass of the joint effect of spirit-master beliefs on cooperation

with distant co-religionists was negative, although also plausibly null (Figure 5; median diff = -0.41, 95% CI = -1.47 to 0.66; from approximately 5.1 tokens when both punishment and omniscience were '0' to 4.7 tokens when both were '1').

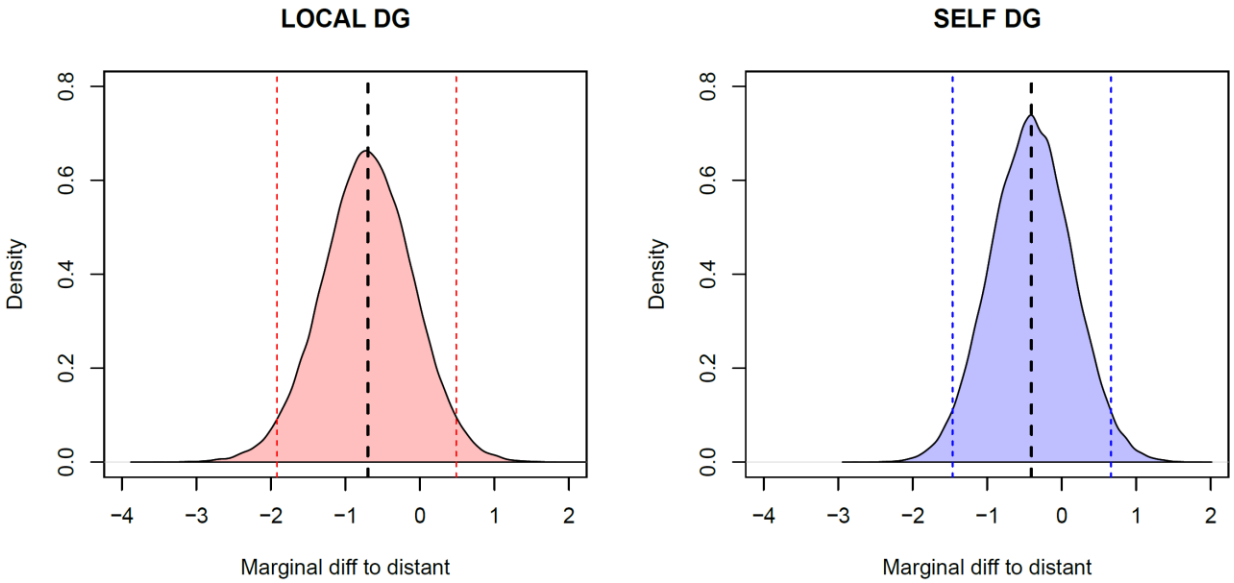


Figure 5: Marginal effect of change in spirit-master punishment and omniscience beliefs on change in cooperation with distant *ovoo* practitioner co-religionist in the Dictator Games (DGs; local DG = left; self DG = right). Dashed black lines denote the median expected difference in cooperation (number of tokens) from both punishment and omniscience taking the value "0" to both as "1," with 95% credible intervals in lighter dashed lines.

3.2.3. Sensitivity Analyses and Complete-Case Results

Although there was some variation in estimates, overall results from both the two sensitivity analyses using imputed data and the complete-case analyses were broadly comparable with the results presented above; that is, most estimates suggested a potential negative association, although also plausibly consistent with there being no relationship (Table 2). While it is difficult to separate the effects of potential selection bias vs. random variation and increased uncertainty in the complete-case results, the broad comparability of these to the imputed results does suggest that any bias due to selection on participants with stronger beliefs in Buddha may have been minimal.

Table 2: Summary of regression results. All estimates are the counter-factual marginal differences in the number of allocations to distant *ovoo* co-religionists in both the Random Allocation Games (RAGs) and Dictator Games (DGs) comparing between when both punishment and omniscience beliefs in spirit-masters take the values "0" vs. "1." These estimates are based on posterior predictions from the corresponding Bayesian model (either aggregate binomial for RAGs, or ordinal models for DGs). All models assume an additive relationship

between punishment and omniscience on cooperation. Covariates in the main analysis include age, sex, education, number of children, material security and game features. Sensitivity analysis 1 includes Buddha punishment and omniscience beliefs as additional covariates. Sensitivity analysis 2 uses single variables as proxies for punishment and omniscience beliefs, rather than mini-scales. Point estimates are median values, with 95% credible intervals in brackets. Results are presented for both the multiple imputation results (MI; $n = 86$) and the complete-case analysis results (CCA; $n = 49$).

	Local RAG		**Self RAG**		**Local DG**		**Self DG**	
	MI	*CCA*	*MI*	*CCA*	*MI*	*CCA*	*MI*	*CCA*
Main analysis	-1.00 (-3.50 to 1.48)	-1.93 (-4.92 to 1.10)	-3.33 (-5.79 to -0.76)	-2.29 (-5.35 to 0.76)	-0.70 (-1.92 to 0.49)	-1.69 (-3.18 to -0.35)	-0.41 (-1.47 to 0.66)	0.06 (-1.10 to 1.22)
Sensitivity analysis 1	-1.60 (-4.30 to 1.07)	-0.95 (-4.74 to 2.71)	-3.22 (-5.87 to -0.45)	-1.95 (-5.61 to 1.74)	-0.64 (-1.92 to 0.59)	-1.51 (-3.10 to -0.16)	-0.19 (-1.31 to 0.91)	-0.65 (-1.90 to 0.55)
Sensitivity analysis 2	-2.13 (-4.98 to 0.77)	-2.41 (-5.84 to 1.35)	-1.78 (-4.64 to 1.10)	-1.22 (-4.86 to 2.41)	-0.49 (-1.83 to 0.78)	-1.92 (-3.63 to -0.45)	-0.33 (-1.47 to 0.80)	-1.04 (-2.24 to 0.12)

4. Discussion

Our results are consistent with punitive and omniscience spirit-master beliefs having a broadly negative – but also plausibly null – relationship with cooperation towards distant co-religionists of the same local religious tradition among the Altai Uriankhai. To the extent these associations can be given a causal interpretation (discussed in more detail below), they suggest that punishment and omniscience beliefs of local spirit-master deities do not expand human cooperation to distant co-religionists in this society, and in fact may potentially *hinder* this expansion.

These patterns of results are in contrast to previous cross-cultural papers which found a positive relationship between supernatural beliefs and cooperation with distant co-religionists, albeit predominantly from more "moralistic" religious traditions (Lang et al., 2019; Purzycki et al., 2016b). However, findings are more consistent when compared against results from similar Inner Asian populations with comparable herding modes of subsistence, environments and religious beliefs and traditions, such as Tyva from southern Siberia (Purzycki & Kulundary, 2018) and Buryat from eastern Mongolia (Ichinkhorloo et al., forthcoming). In these populations there was also suggestive evidence for a negative relationship between spirit-master beliefs and donations to distant co-religionists (Buddhists in Tyva; *ovoo* practitioners in Buryat). Compared to game behaviour from other populations (Lang et al., 2023; Purzycki et al., 2024), cooperation in these Inner Asia sites was also largely fair and unbiased, albeit with slightly more donations to self and local co-religionists in Altai Uriankhai compared to Tyva and Buryat (see also Berniūnas et al., 2020 for high levels of DG cooperation in a different Mongolian sample from Ulaanbaatar).

Rather than promoting cooperation with distant co-religionists, taken together these results suggest that beliefs in spirit-masters may in fact potentially promote parochial cooperation. On the one hand, this is counter to general expectations of the "supernatural punishment" hypothesis that these punitive and omniscience beliefs would promote cooperation (Johnson & Bering, 2006; Johnson & Krüger, 2004), especially with distant co-religionists (Norenzayan et al., 2016). With the specific ethnographic context in mind, however, this pattern is perhaps less surprising. For instance, spirit-masters tend to have very local and geographically-bounded concerns, such as resource management and mediating territorial boundaries (Purzycki, 2010). The extent to which spirit-master beliefs would promote generalised moralistic behaviour is therefore equivocal, while this local focus may facilitate cooperation among *local* co-religionists at the expense of *distant* ones (although why they might promote selfishness – as per the self RAG condition – is unclear). However, when interpreting these results, we must bear in mind that although the majority of models suggested a negative relationship, most were also consistent with there being no causal effect.

A somewhat unexpected auxiliary finding was that spirit-masters were generally rated as more punitive and omniscient than Buddha, especially in the imputed data which included some

participants who responded that they did not believe in Buddha and hence responded "don't know" to many of these questions. From previous work among Tyvans (Purzycki & Kulundary, 2018) and Buryats (Ichinkhorloo et al., forthcoming), punishment and omniscience beliefs towards Buddha and spirit-masters were largely comparable; it is uncertain why beliefs about Buddha would be weaker, and spirit-master beliefs stronger, among the Altai Uriankhai, although a few reasons are possible (none of which are mutually-exclusive). First, the influence of Buddhism on everyday life may be weaker here compared to other places in Inner Asia; for instance, among Altai Uriankhais, Buddhism does not necessarily involve strong institutional connections to the Buddhist church, and most local people do not visit monasteries or lamas regularly. Second, the local climate and subsistence methods means living conditions are arguably harsher for the Altai Uriankhai in Duut *sum* compared to Tyvans from Kyzyl city and Buryat from Bayandun *sum*; for instance, they are at greater risk of *zuds* (extreme winter weather conditions), live at higher elevations, have less annual precipitation, and are more exposed to wind. These factors, combined with a predominantly herding population without central heating, mean that ecological challenges relating to making a living and environmental productivity may be more pronounced among the Altai Uriankhai; as spirit-masters are primarily concerned with similar factors relating to the local ecology – such as resource management and mediating between-group dynamics – the physical environment may make these spirit-master beliefs more salient among the Altai Uriankhai compared to other places in Inner Asia. Finally, compared to city-dwelling Tyvans and the Mongolian Buryat population, the Altai Uriankhai community have lived in their present area for considerably longer (since the 17th or 18th century, compared to the period following the Russian October Revolution for Buryats in Bayandun *sum*; Shimamura, 2004); given this, their symbolic connections to their homeland may be more traditional and intimate, potentially reflected in stronger beliefs towards local spirit-masters than Buddhist religious traditions. However, these suggestions are necessarily tentative, and more detailed comparative work is needed to understand reasons for variation in religious beliefs and practices between these Inner Asian sites.

This research has several strengths, including: i) ethnographically-informed research design and data collection, tailored to the specific local context and aiding interpretation of results; ii) use of formal causal inference methods to make our estimand – and assumptions needed to estimate said estimand without bias – unambiguous; iii) a clear theoretically-informed research question with analyses pre-registered, lowering the risk of questionable research practices biasing results; and iv) data collection as part of a larger cross-cultural project, providing a unique resource of religiosity and cooperation in 22 small-scale societies.

Of course, there are also a number of limitations of this work and ways in which future research can improve upon it. First, any causal interpretation from observational data rests on very strong – and largely untestable – assumptions. Three key assumptions are no confounding bias, no selection bias, and no measurement bias (Hernán & Robins, 2020). Regarding confounding bias,

we aimed to include all relevant confounders in our causal and statistical models (Figure 1), including different assumptions regarding Buddha beliefs, which may plausibly be a confounder and/or a mediator. When exploring these different assumptions, there was little overall difference in interpretation of results, suggesting that our results are somewhat robust to these different causal assumptions. However, we cannot rule out potential unmeasured confounding by other factors we did not have data on (e.g., wealth, personality, cognition). Longitudinal data collection would have also provided stronger evidence for a causal interpretation by clearly delineating confounders vs. mediators and removing the risk of reverse causality between religious beliefs and cooperation (i.e., prior cooperative dispositions impacting both supernatural beliefs and measured cooperation). Regarding selection bias, our complete-case analytic sample was biased towards participants who believed in Buddha; however, the comparability of the imputed and complete-case results suggests that missing data may not have resulted in substantial bias. Our sample also appears broadly representative of the wider Altai Uriankhai population, lowering the risk of selection bias due to sample-population mismatch. Finally, for a causal interpretation we have to assume no measurement bias. This assumption is difficult to verify objectively, especially as beliefs are subjective, self-reported and may be biased by numerous factors (e.g., social desirability, demand characteristics, being forced to condense – perhaps previously implicit – beliefs into a "yes/no" response). However, the fieldworker (AM) has extensive experience and knowledge working with this population, hopefully minimising these risks. Data are also broadly consistent with known ethnographic data, and data from other similar sites (i.e., Tyva and Buryat), providing a further degree of assurance. In sum, while we cannot claim a definitive causal interpretation, there are good reasons to believe that any potential bias may be rather small. At the very least, this is the best evidence we have for a causal interpretation, at present.

A further limitation is that the sample size was relatively small, meaning that our study may have been underpowered to detect small – but still potentially relevant, especially when scaled up to the population level – effect sizes. While *a priori* power analyses indicated a sample size of 40 per condition would be sufficient to reliably detect a 4%-point difference in RAG cooperation with 90% power (i.e., from 50% to 54%; approximately a one-token difference in cooperation), many factors can impact power, such as the distribution of supernatural belief variables, variation in cooperative behaviour, and the inclusion of confounding factors. As such, larger samples are required to distinguish between negative vs. null effect sizes. An additional limitation is that this paper focused only on cooperation within "local" religious traditions, meaning the effect of beliefs regarding "moralistic" tradition types (i.e., Buddha) on cooperation towards distant Buddhists in this population is unknown. That said, comparable research in neighbouring Tyva found little evidence that Buddha beliefs impacted cooperation with distant Buddhist co-religionists (Purzycki & Kulundary, 2018), while work in Benin has explored both of these streams in the same population (Dalla Ragione, forthcoming).

In summary, punishment and omniscience beliefs regarding local spirit-master deities do not promote cooperation with distant co-religionists among the Altai Uriankhai. If anything, these beliefs appear to promote *parochial* cooperation and selfishness, although a null relationship is also plausible. While the interpretation of these results is ambiguous, they do suggest that – in this Mongolian population at least – beliefs regarding *local* supernatural agents do not expand the circle of human cooperation to other distant co-religionists of the same local religious tradition.

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Disclosure statement
The authors declare no conflicts of interest.

Authors' roles
AM collected data, DM-S and BGP managed the project and contributed to analysis, and BGP designed the protocols. All authors contributed to writing this manuscript.

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Data availability statement
All data and analysis code are available on the project GitHub website: https://github.com/bgpurzycki/GGSL-Project/tree/main/Code/Objective%201/Altai%20Uriankhai.

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